DPT Exercise 10

github.com/aiulus/dpt-ss25

START: 12:15

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1) Maximum Likelihood Estimation (MLE)
   - w/ Exp(λ)
   - w/ arbitrary pdf
   - properties of estimators (unbiasedness)
2) Auxiliary:
   - Jensen's inequality
   - Gamma distribution
3) Recap: Constructing statistical models (probability spaces)
   Combinatorics
   verifying a given pdf

DPT Exercises

Exercise sheet 10 — Niki Kilbertus

Exercise 1 ([repeated from sheet 9:] mle for uniform distribution)**.** Let $X_1, \dots, X_n$ be iid RVRVs from a uniform distribution $\mathrm{Unif}(0, \theta)$ with unknown parameter $\theta > 0$. The pdf of X_i is $p(x_i \mid \theta) = \frac{1}{\theta}\chi_{[0,\theta]}(x_i)$.

(i) Write down the likelihood function $L(\theta \mid x)$ for an observed sample $x = (x_1, \dots, x_n)$. Be careful with the indicator functions.

(ii) Find the maximum likelihood estimator (mle) $\hat{\theta}_{\mathrm{mle}}$ for θ.
[Hint: The likelihood function may not be differentiable with respect to θ, but the maximum value can be found by reasoning about monotonicity and where it is non-zero.]

(iii) Calculate the expected value $\mathbb{E}[\hat{\theta}_{\mathrm{mle}}]$. Is $\hat{\theta}_{\mathrm{mle}}$ an unbiased estimator for θ?
[Hint: First find the cdf of $\hat{\theta}_{\mathrm{mle}}$, then its pdf, and then compute the expectation.]

Exercise 2 (mle for exponential distribution)**.** Let $X_1, \dots, X_n$ be iid RVRVs from an exponential distribution for $n \in \mathbb{N}_{>1}$ with unknown rate parameter $\lambda > 0$, i.e., $X \sim (\mathrm{Exp}(\lambda))_{\lambda \in \mathbb{R}_{>0}}$.

(i) Write down the likelihood function $L(\lambda \mid x)$ and the log-likelihood function $\ell(\lambda \mid x)$ for an observed sample $x = (x_1, \dots, x_n)$ when given only the pdf of an $\mathrm{Exp}(\lambda)$ distribution.

(ii) Find the maximum likelihood estimator $\hat{\lambda}_{\mathrm{mle}}$ for λ.

(iii) Calculate the expected value $\mathbb{E}[\hat{\lambda}_{\mathrm{mle}}]$. Is $\hat{\lambda}_{\mathrm{mle}}$ an unbiased estimator for λ?
[Hint: Since $X_i \sim \mathrm{Exp}(\lambda) = \Gamma(1, \lambda)$, you can use that $S_n = \sum_{i=1}^n X_i \sim \Gamma(n, \lambda)$. Recall that $1/\bar{X}_n = n/S_n$ and use that for $Y \sim \Gamma(\alpha, \beta)$ we have that $\mathbb{E}[1/Y] = \beta/(\alpha - 1)$ if $\alpha > 1$.]

Exercise 3. Let X be a RVRV with pdf

$$p_X(x) = \begin{cases} 2c^2/x^3 & \text{if } x \geq c\,, \\ 0 & \text{otherwise}\,, \end{cases}$$

with $c > 0$.

(i) Show that p_X is a valid pdf for all $c > 0$.

(ii) Determine the cdf of X.

(iii) Let $X_1, \dots, X_n$ be iid RVRVs from this distribution. Find the mle for c.
[Hint: For the likelihood function, consider the case where a sample value is less than c.]

Exercise 4. You're throwing $n \in \mathbb{N}$ darts at a target. Let $(X_i)_{i \in [n]}$ be iid random variables with unknown mean μ and unknown standard deviation σ, representing the distance of the i-th arrow from the center of the target. Furthermore, let $\hat{\mu} = \sum_{i=1}^n \lambda_i X_i$ be a convex combination of the individual X_i (i.e., $\lambda_i \geq 0$ for all $i \in [n]$ and $\sum_{i=1}^n \lambda_1 = 1$).

(i) Show that $\hat{\mu}$ is an unbiased estimator for μ.

(ii) Prove that $\frac{1}{n}\sum_{i=1}^n \lambda_i^2 \geq \frac{1}{n^2}$.
[Hint: You can do this quickly using Jensen's inequality for an appropriate discrete random variable.]

(iii) Determine the coefficients λ_i that minimize the mse of the estimator $\hat{\mu}$.

Exercise 5. You have a box with $n \in \mathbb{N}_{>1}$ toasters among which a fixed number $d \in \{0, \dots, d\}$ is broken. You draw two toasters uniformly at random from the box.

(i) What is the probability distribution P_d (depending on d) of the number of broken toasters among the two selected ones?
[Hint: Don't overcomplicate it, this is going to be a distribution over a set with just 3 elements. Feel free to specify the distribution via it's pmf.]

(ii) Assume d to be unknown and we want to estimate it from a (uniform) sample of two toasters. Write down an adequate statistical model and briefly explain it.

(iii) Provide an unbiased estimator for d in this model and justify your choice.

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(i) Write down the likelihood function $L(\theta \mid x)$ for an observed sample $x = (x_1, \dots, x_n)$. Be careful with the indicator functions.

> **Definition 3.13** (likelihood function). Let $(X_1, \dots, X_n)$ be a random sample from a statistical model $(\Omega, \mathcal{A}, P_\Theta)$ where all $P_\theta \in P_\Theta$ have a pmf or density $p_\theta(\cdot) =: p(\cdot \mid \theta)$. Given observed data $x = (x_1, \dots, x_n)$, the *likelihood (function)* is the following function
>
> $$L(\cdot \mid x) : \Theta \to \mathbb{R}_{\geq 0} : \theta \mapsto L(\theta \mid x) := p_\theta(x) = p(x \mid \theta) = p(x_1, \dots, x_n \mid \theta) .$$
>
> If $X_1, \dots, X_n$ are iid, then $L(\theta \mid x) = \prod_{i=1}^n p(x_i \mid \theta)$. The *log-likelihood function* is $\ell(\theta \mid x) := \ln(L(\theta \mid x))$.

(i) $L(\theta \mid \vec{x}) = \prod_{i=1}^n p(x_i \mid \theta) = \prod_{i=1}^n \frac{1}{\theta} \cdot \chi_{[0,\theta]}(x_i) = \frac{1}{\theta^n} \cdot \prod_{i=1}^n \chi_{[0,\theta]}(x_i)$

$= \frac{1}{\theta^n} \cdot \begin{cases} 1 & \text{if } \theta \geq \max\{x_1, \dots, x_n\} \\ 0 & \text{otherwise} \end{cases}$ $\quad := x_{\max}$

$= \frac{1}{\theta^n} \cdot \chi_{[x_{\max}, \infty)}(\theta)$

$= \begin{cases} 1 & \text{if all } x_i \in [0, \theta] \\ 0 & \text{otherwise} \end{cases}$

$= \begin{cases} 1 & \text{if } \theta \geq \min\{x_{1:n}\} \geq \max\{x_{1:n}\} \geq \theta \\ 0 & \text{otherwise} \end{cases}$

MLE in general:

- $L(\theta \mid \vec{x}) = P(X_1 = x_1, \dots, X_n = x_n \mid \theta)$
  $\overset{\text{i.i.d.}}{=} \prod_{j=1}^n P(X_j = x_j \mid \theta) = \prod_{j=1}^n p_{X_j}(x_j ; \theta)$ parametric pdf.

Steps:

1) Write down $L(\theta \mid \vec{x})$ (and $\ell(\theta \mid \vec{x}) = \ln(L(\theta \mid \vec{x}))$)

2) Compute $\hat{\theta}_{MLE} = \underset{\theta}{\operatorname{argmax}} \, L(\theta \mid \vec{x}) = \underset{\theta}{\operatorname{argmax}} \, \ell(\theta \mid \vec{x})$

by solving for θ in:

- $\frac{\partial}{\partial \theta} \ell(\theta \mid \vec{x}) = 0$

(and optionally verifying $\frac{\partial^2}{\partial \theta^2} \ell(\theta \mid \vec{x}) < 0$)

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(ii) Find the maximum likelihood estimator (mle) $\hat{\theta}_{\mathrm{mle}}$ for θ.
[Hint: The likelihood function may not be differentiable with respect to θ, but the maximum value can be found by reasoning about monotonicity and where it is non-zero.]

> **Definition 3.27** (maximum likelihood estimation (mle)). The *maximum likelihood estimator (mle)* $\hat{\theta}_{\mathrm{mle}}$ is the value of $\theta \in \Theta$ that maximizes the likelihood function $L(\theta \mid x)$, i.e.,
>
> $$\hat{\theta}_{\mathrm{mle}}(x) \in \arg\max_{\theta \in \Theta} L(\theta \mid x) = \arg\max_{\theta \in \Theta} \ell(\theta \mid x) \,.$$
>
> Maximizing the log-likelihood is often mathematically simpler. If $\ell(\theta \mid x)$ is differentiable and the maximum is in the interior of Θ, $\hat{\theta}_{\mathrm{mle}}$ can often be found by solving the *likelihood equation* $\nabla_\theta \ell(\theta \mid x) = 0$ for θ.

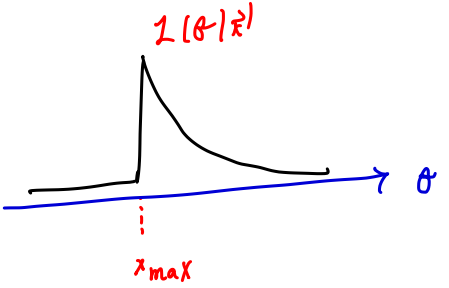
unless difficult-to-evaluate $\hat{\theta}_{MLE}$, use

1) monotonicity

2) plot the function

(ii) $L(\theta \mid \vec{x}) = \frac{1}{\theta^n} \cdot \chi_{[x_{max}, \infty)}(\theta)$

$\hat{\theta}_{MLE} = \arg\max_{\theta} \begin{cases} 1/\theta^n & \text{if } \theta \geq x_{max} \\ 0 & \text{otherwise} \end{cases}$



$= \max\{X_1, \dots, X_n\}$

→ be careful to use the RVRV's here

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(iii) Calculate the expected value $\mathbb{E}[\hat{\theta}_{\mathrm{mle}}]$. Is $\hat{\theta}_{\mathrm{mle}}$ an unbiased estimator for θ?
[Hint: First find the cdf of $\hat{\theta}_{\mathrm{mle}}$, then its pdf, and then compute the expectation.]

Definition 3.20 (properties of estimators). Let $\hat{\theta}$ be an estimator for the parameter θ. The quality of this estimator can be assessed according to various properties.

(i) **Bias:** The *bias* of an estimator $\hat{\theta}$ is $\mathrm{bias}[\hat{\theta}] = \mathbb{E}[\hat{\theta}] - \theta$. An estimator is *unbiased* if $\mathrm{bias}[\hat{\theta}] = 0$, i.e., $\mathbb{E}[\hat{\theta}] = \theta$ for all $\theta \in \Theta$. (What is the expectation over?)

(iii) $\hat{\theta}_{MLE} = \max\{X_1, \dots, X_n\}$; check; $\mathbb{E}[\hat{\theta}_{MLE}] = \theta$

$$\mathbb{E}[\hat{\theta}_{MLE}] = \int_{-\infty}^{\infty} x \cdot f_{\hat{\theta}_{MLE}}(x)\, dx = \int_{-\infty}^{\infty} x \cdot \frac{n}{\theta^n} \cdot x^{n-1} \cdot \chi_{[0,\theta]}(x)\, dx$$

$$= \frac{n}{\theta^n} \cdot \int_0^{\theta} x^n\, dx = \frac{n}{\theta^n} \cdot \left[\frac{1}{n+1} \cdot x^{n+1} \right]_0^{\theta}$$

$$= \frac{n}{n+1} \cdot \frac{1}{\theta^n} \cdot \theta^{n+1} = \frac{n}{n+1} \cdot \theta$$

consistent but biased

biased; $n = 2$ $\Rightarrow \frac{2}{3} \cdot \theta$

consistent; $\lim_{n \to \infty} \dots = \theta$

Datenblatt:

$$F_{\hat{\theta}_{MLE}}(x) = P(\hat{\theta}_{MLE} \leq x)$$

$$= P(\max\{X_1, \dots, X_n\} \leq x)$$

$$= P(X_1 \leq x, \dots, X_n \leq x) = \prod_{i=1}^{n} P(X_i \leq x)$$

» $X_i \sim \mathrm{Unif}([0, \theta])$

$$\Rightarrow F_{X_i}(x) = \begin{cases} 0 & \text{if } x < 0 \\ x/\theta & \text{if } x \in [0, \theta] \\ 1 & \text{if } x > \theta \end{cases}$$

$$= \prod_{i=1}^{n} x/\theta \cdot \chi_{[0,\theta]}(x) = x^n/\theta^n \cdot \chi_{[0,\theta]}(x)$$

$$\Rightarrow p_{\hat{\theta}_{MLE}}(x) = \frac{\partial}{\partial x} \frac{x^n}{\theta^n} \cdot \chi_{[0,\theta]}(x)$$

$$= \frac{n}{\theta^n} \cdot x^{n-1} \cdot \chi_{[0,\theta]}(x)$$

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with $c > 0$.

(i) Show that p_X is a valid pdf for all $c > 0$.

(ii) Determine the cdf of X.

(ii) $F_X(x) = \int_{-\infty}^{x} p_X(t)\,dt = \int_{-\infty}^{x} 2c^2 \cdot t^{-3} \cdot \chi_{[c,\infty)}(t)\,dt$

$\max(-\infty, c)$

$= 2c^2 \cdot \int_c^x t^{-3}\,dt = 2c^2 \cdot \left[-\frac{1}{2} \cdot t^{-2}\right]_c^x$

$= -c^2 \cdot \left(x^{-2} - c^{-2}\right) = \left(1 - c^2/x^2\right) \cdot \chi_{[c,\infty)}(x)$

(i) $\int_{-\infty}^{\infty} p_X(x)\,dx \overset{?}{=} 1$

$\int_{-\infty}^{\infty} 2c^2\, x^{-3} \cdot \chi_{[c,\infty)}(x)\,dx = 2c^2 \cdot \int_c^{\infty} x^{-3}\,dx$

$= 2c^2 \cdot \left[-\frac{1}{2} x^{-2}\right]_c^{\infty} = -c^2 \cdot \left[\lim_{x\to\infty} x^{-2} - c^{-2}\right]$

$\lim_{x\to\infty} x^{-2} = 0$

$= -c^2 \cdot (-c^{-2}) = 1$ ☐

Exercise 3. Let X be a RVRV with pdf

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with $c > 0$.

(iii) Let $X_1, \ldots, X_n$ be iid RVRVs from this distribution. Find the mle for c.
[Hint: For the likelihood function, consider the case where a sample value is less than c.]

$= \begin{cases} 1 & \text{if all } x_i \geq c \\ 0 & \text{otherwise} \end{cases} = \begin{cases} 1 & \text{if } x_{min} \geq c \\ 0 & \end{cases}$

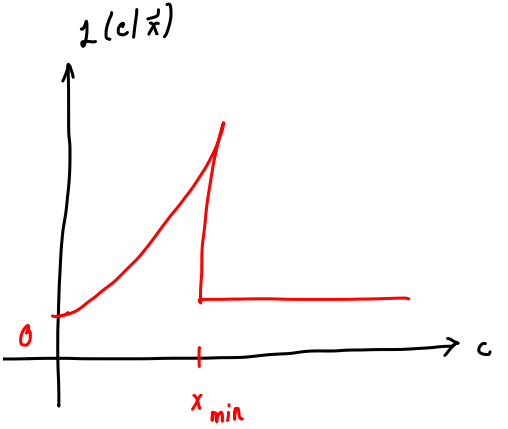
$= \chi_{(-\infty, x_{min}]}(c)$

1) $L(c|\vec{x}) = \prod_{i=1}^{n} p_{X_i}(x_i; c) = \prod_{i=1}^{n} 2c^2 \cdot x_i^{-3} \cdot \chi_{[c,\infty)}(x_i) = 2^n c^{2n} \cdot \left(\prod_{i=1}^{n} x_i^{-3}\right) \cdot \prod_{i=1}^{n} \chi_{[c,\infty)}(x_i)$

$= 2^n c^{2n} \cdot \left(\prod_{i=1}^{n} x_i^{-3}\right) \cdot \chi_{(-\infty, x_{min}]}(c)$

2) $\hat{c}_{MLE} = \underset{c}{\arg\max} \begin{cases} 2^n c^{2n} \cdot \left(\prod_{i=1}^{n} x_i^{-3}\right) & \text{if } c \leq x_{min} \\ 0 & \text{otherwise} \end{cases} = x_{min}$

cont.!



$\Rightarrow \hat{c}_{MLE} = \min\{X_1, \ldots X_n\}$

random variables!

Exercise 4. You're throwing $n \in \mathbb{N}$ darts at a target. Let $(X_i)_{i\in[n]}$ be iid random variables with unknown mean μ and unknown standard deviation σ, representing the distance of the i-th arrow from the center of the target. Furthermore, let $\hat{\mu} = \sum_{i=1}^{n} \lambda_i X_i$ be a convex combination of the individual X_i (i.e., $\lambda_i \geq 0$ for all $i \in [n]$ and $\sum_{i=1}^{n} \lambda_1 = 1$).

(i) Show that $\hat{\mu}$ is an unbiased estimator for μ. $\mathbb{E}[\hat{\mu}] = \mu$

$$\mathbb{E}[\hat{\mu}] = \mathbb{E}\left[\sum_{i=1}^{n} \lambda_i \cdot X_i\right] = \sum_{i=1}^{n} \lambda_i \cdot \mathbb{E}[X_i] = \sum_{i=1}^{n} \lambda_i \cdot \mu = \mu \cdot \sum_{i=1}^{n} \lambda_i = \mu \quad \square$$

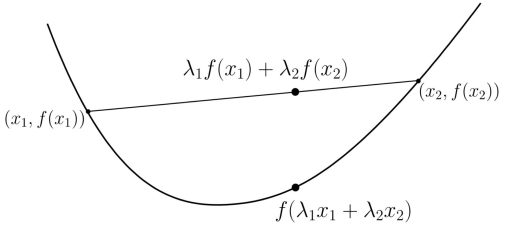
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(ii) Prove that $\frac{1}{n}\sum_{i=1}^n \lambda_i^2 \geq \frac{1}{n^2}$.
[Hint: You can do this quickly using Jensen's inequality for an appropriate discrete random variable.]

Theorem 2.42 (Jensen's Inequality). *Let X be an RVRV such that $\mathbb{E}[X]$ is finite. Let $g : \mathbb{R} \to \mathbb{R}$ be a convex function, that is for any $x, y \in \mathbb{R}$ and $t \in [0,1]$ we have $f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2)$. Then*

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])\,.$$

If g is strictly convex, then equality holds if and only if $X = \mathbb{E}[X]$ with probability 1 (i.e., X is a constant). If g is concave, the inequality is reversed, $\mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$.



$t \sim \frac{1}{n} \cdot \sum_{i=1}^n \lambda_i^2 \geq \frac{1}{n^2}$

$\sim$ uniform over $\{\lambda_1, \dots \lambda_n\}$, so

$$p_\sim(\lambda) = \begin{cases} \frac{1}{n} & \text{if } \lambda \in \{\lambda_1, \dots \lambda_n\} \\ 0 & \text{otherwise} \end{cases}$$

$g(x) = x^2$ convex

$$\mathbb{E}[Z^2] = \sum_{\lambda \in \{\lambda_i\}_{i=1}^n} \lambda^2 \cdot \frac{1}{n} = \sum_{i=1}^n \lambda_i^2 \cdot \frac{1}{n}$$

$$\geq (\mathbb{E}[Z])^2 = \left(\sum_{i=1}^n \lambda_i \frac{1}{n}\right)^2$$

$$= \left(\frac{1}{n} \cdot \sum_{i=1}^n \lambda_i\right)^2 = 1/n^2$$

$= 1$

$\uparrow$ Jensen's inequality

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(iii) Determine the coefficients λ_i that minimize the mse of the estimator $\hat{\mu}$.

(1) $\mathrm{Var}[X+Y] = \mathrm{Var}[X] + \mathrm{Var}[Y]$ if $X \perp\!\!\!\perp Y$

$\hat{\mu}$ unbiased $\Rightarrow$ $\mathrm{mse}[\hat{\mu}] = \mathrm{Var}[\hat{\mu}] = \mathrm{Var}\left[\sum_{i=1}^{n} \lambda_i \cdot X_i\right] = \sum_{i=1}^{n} \mathrm{Var}[\lambda_i \cdot X_i] = \sum_{i=1}^{n} \lambda_i^2 \cdot \mathrm{Var}[X_i] = \sigma^2 \cdot \sum_{i=1}^{n} \lambda_i^2 \geq \sigma^2/n$ (*)

($\mathrm{Var}[X_i] = \sigma^2$; $\sum_{i=1}^{n} \lambda_i^2 \geq 1/n$)

(*) from (ii): $\frac{1}{n} \cdot \sum_{i=1}^{n} \lambda_i^2 \geq \frac{1}{n^2}$

$\lambda_{\min} = \underset{\{\lambda_i\}_{i=1}^{n}}{\arg\min}\ \mathrm{mse}[\hat{\mu}] = \sigma^2 \cdot \sum_{i=1}^{n} \lambda_i^2 \geq \sigma^2/n$

(not unique) solutions: $\lambda_i = 1/n \quad \forall i \in [n]$

Exercise 5. You have a box with $n \in \mathbb{N}_{>1}$ toasters among which a fixed number $d \in \{0, \dots, d\}$ is broken. You draw two toasters uniformly at random from the box.

(i) What is the probability distribution P_d (depending on d) of the number of broken toasters among the two selected ones?

[Hint: Don't overcomplicate it, this is going to be a distribution over a set with just 3 elements. Feel free to specify the distribution via it's pmf.]

$\Omega = \{0,1,2\}$; $\mathcal{A} = \mathcal{P}(\{0,1,2\})$; $P_d(\omega) = \begin{cases} _ & \text{if } \omega = 0 \\ _ & \text{if } \omega = 1 \\ _ & \text{if } \omega = 2 \\ 0 & \text{otherwise} \end{cases}$ $\longrightarrow \dfrac{\binom{n-d}{2}}{\binom{n}{2}} = \dfrac{(n-d)(n-d-1)}{n \cdot (n-1)}$

obs: Hypergeometric distr.

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(ii) Assume d to be unknown and we want to estimate it from a (uniform) sample of two toasters. Write down an adequate statistical model and briefly explain it.

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Definition 3.20 (properties of estimators). Let $\hat{\theta}$ be an estimator for the parameter θ. The quality of this estimator can be assessed according to various properties.

(i) **Bias:** The *bias* of an estimator $\hat{\theta}$ is $\text{bias}[\hat{\theta}] = \mathbb{E}[\hat{\theta}] - \theta$. An estimator is *unbiased* if $\text{bias}[\hat{\theta}] = 0$, i.e., $\mathbb{E}[\hat{\theta}] = \theta$ for all $\theta \in \Theta$. (What is the expectation over?)

(ii) **Consistency:** An estimator $\hat{\theta}_n$ is *consistent* if $\hat{\theta}_n \xrightarrow{P} \theta$. It is *strongly consistent* if $\hat{\theta}_n \xrightarrow{a.s.} \theta$.

(iii) **Variance:** The variance of an estimator is $\text{var}[\hat{\theta}] = \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]$. For multi-variate θ, this turns into $\text{cov}[\hat{\theta}, \hat{\theta}]$.

Definition 3.13 (likelihood function). Let $(X_1, \ldots, X_n)$ be a random sample from a statistical model $(\Omega, \mathcal{A}, P_\Theta)$ where all $P_\theta \in P_\Theta$ have a pmf or density $p_\theta(\cdot) =: p(\cdot \mid \theta)$. Given observed data $x = (x_1, \ldots, x_n)$, the *likelihood (function)* is the following function

$$L(\cdot \mid x) : \Theta \to \mathbb{R}_{\geq 0} : \theta \mapsto L(\theta \mid x) := p_\theta(x) = p(x \mid \theta) = p(x_1, \ldots, x_n \mid \theta) \,.$$

If $X_1, \ldots, X_n$ are iid, then $L(\theta \mid x) = \prod_{i=1}^n p(x_i \mid \theta)$. The *log-likelihood function* is $\ell(\theta \mid x) := \ln(L(\theta \mid x))$.

Definition 3.27 (maximum likelihood estimation (mle)). The *maximum likelihood estimator (mle)* $\hat{\theta}_{\text{mle}}$ is the value of $\theta \in \Theta$ that maximizes the likelihood function $L(\theta \mid x)$, i.e.,

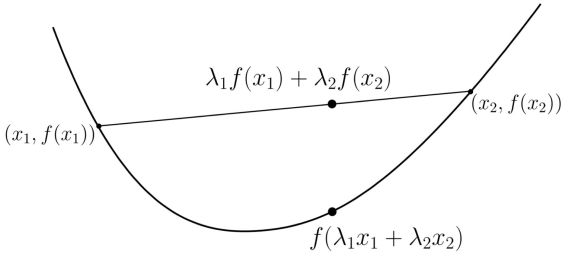
$$\hat{\theta}_{\text{mle}}(x) \in \arg\max_{\theta \in \Theta} L(\theta \mid x) = \arg\max_{\theta \in \Theta} \ell(\theta \mid x) \,.$$

Maximizing the log-likelihood is often mathematically simpler. If $\ell(\theta \mid x)$ is differentiable and the maximum is in the interior of Θ, $\hat{\theta}_{\text{mle}}$ can often be found by solving the *likelihood equation* $\nabla_\theta \ell(\theta \mid x) = 0$ for θ.

Theorem 2.42 (Jensen's Inequality). *Let X be an RVRV such that $\mathbb{E}[X]$ is finite. Let $g : \mathbb{R} \to \mathbb{R}$ be a convex function, that is for any $x, y \in \mathbb{R}$ and $t \in [0,1]$ we have $f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2)$. Then*

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])\,.$$

If g is strictly convex, then equality holds if and only if $X = \mathbb{E}[X]$ with probability 1 (i.e., X is a constant). If g is concave, the inequality is reversed, $\mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$.



(iii) *Gamma distribution:* For $\alpha > 0$, $\beta > 0$, the $\Gamma(\alpha, \beta)$ distribution on $\Omega = \mathbb{R}_{>0} = (0, \infty)$ is characterized via the pdf

$$p_{\Gamma(\alpha,\beta)} : \mathbb{R} \to \mathbb{R}_{\geq 0},\ x \mapsto \chi_\Omega(x)\, \frac{\beta^\alpha}{\Gamma(\alpha)}\, x^{\alpha-1} e^{-\beta x}\ .$$